

Tusher Bhomik

+8801690090528 | tusherbhomik@gmail.com | Palashi, Newmarket, Dhaka-1205

Portfolio: [tusherbhomik.github.io](https://github.com/tusherbhomik) | LinkedIn: [tusher-bhomik](#) | GitHub: [Tusherbhomik](#) | LeetCode: [tusher_bhomik](#)

EDUCATION

- **Bangladesh University of Engineering and Technology** January 2022 – June 2026, CGPA: 3.3164/4.00
Bachelor of Science in Computer Science & Engineering Dhaka, Bangladesh
- **Notre Dame College** 2018 – 2020, GPA: 5.00/5.00
Higher Secondary Certificate (HSC), Science Dhaka, Bangladesh

RELEVANT COURSEWORK

- Machine Learning, Artificial Intelligence, Data Structures, Algorithms, Theory of Computation, Computer Architecture, Operating Systems, Compiler Design, Database Management Systems, Computer Networks, Software Engineering, Computer Graphics

RESEARCH

- **Age-Aware Neurological Gait Pattern Discovery from Biomechanical Signals: Multi-Phase Stroke Severity Analysis** 2025 – 2026
Undergraduate Thesis (defended) | Supervisor: Mahmuda Naznin, Dept. of CSE, BUET Dhaka, Bangladesh
 - An **age-normative framework** for uncovering neurological relations from biomechanical gait features using **machine learning with explainable AI**.
- **Reinforcement Learning for Reliable and Security-Aware LLM Agents (Ongoing)** 2026 – Present
Supervisor: Md. Rizwan Parvez, Qatar Computing Research Institute Doha, Qatar
 - Investigating reinforcement learning-based optimization to enhance the reliability and accuracy of function calling behavior in large language models for complex tool-use scenarios.
 - Exploring reinforcement learning techniques for developing security-aware agentic systems capable of secure reasoning, vulnerability analysis, and autonomous decision-making in software repositories.

PROJECTS

- **HealthSync** 2025
Spring Boot, React, PostgreSQL, Firebase, Docker, Azure, CI/CD [GitHub](#)
 - Developed a comprehensive healthcare platform connecting patients and doctors with appointment scheduling, digital prescription management, and medical history tracking.
 - **Containerized** the services with **Docker** and deployed to **Microsoft Azure** through a **CI/CD pipeline**.
- **Arxiv Research Paper RAG System** 2026
Python, SQL, SQLite, FastAPI, ChromaDB, Data Analysis [GitHub](#)
 - Built an end-to-end **data pipeline** for ingesting, cleaning, and analyzing arXiv research paper data using Python and SQL.
 - Developed a **RAG-based API system** using FastAPI and ChromaDB for semantic retrieval and question-answering.
- **Evidence-Grounded Multilingual Image-Edit Evaluator** 2026
Python, VLMs (Qwen, Gemma), Multilingual Benchmark
 - Built a multilingual benchmark (~6K image-edit pairs from **Pico-Banana-400K**, English/Bangla/Hindi) for instruction-following image editing.
 - Designed an **evidence-grounded multi-label evaluator** fusing visual and textual features to diagnose **11 fine-grained failure categories** and benchmarked against open-source VLM judges (Qwen, Gemma).
- **Enhanced xv6-riscv Kernel** 2024
C, Assembly, Makefile [GitHub](#)
 - Implemented an **MLFQ scheduler** with aging and added **kernel-level threading** support.
 - Added custom system calls (trace, sysinfo) and a Bash autograder for testing.
- **Mini C Compiler** 2024
CSE-310 Compiler Design, C, 8086 Assembly [GitHub](#)
 - Built a **mini C compiler** from scratch in four phases: Symbol Table, Lexical Analysis, Semantic Analysis, and Intermediate Code Generation.
 - Generated **8086 assembly** as intermediate output to validate end-to-end compilation workflow.

ACHIEVEMENTS

- **5th Place — Tech Career Hunt 2.0**, organized by Shohoj Coding. May 2026
- **Selected for AI/ML Internship — Save the Children**, through their Hackathon (AI/ML Track).

SKILLS

- **Research Interests:** LLMs, Reinforcement Learning, Trustworthy AI, AI Security, Software Security, Systems Security, Operating Systems, Human-Computer Interaction, Cyber-Physical Systems
- **Programming:** Python, C++, C, Java, JavaScript
- **ML/AI:** PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV, LLMs, vLLM
- **Backend:** FastAPI, Spring Boot, Express.js
- **Frontend:** React, Next.js, Material-UI
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Tools:** Git, Docker, Linux

ADDITIONAL INFORMATION

Languages: English, Bengali. **Interests:** Photography, Traveling, Badminton, Cycling.

REFERENCES

- **Prof. Dr. Mahmuda Naznin** — Thesis Supervisor
Department of CSE, Bangladesh University of Engineering and Technology
Email: mahmudanaznin@gmail.com